

Supplementary Materials

This appendix collects the granular tables and figures extracted from the main text to keep the narrative focused on the taxonomy, its empirical impact and its validation. Each is reproducible from the released parquet and the audit scripts.

Table S1: A4 sensitivity sweep on the released v1.0.1 parquet (46,307 stations, 123 systems). Each row reports the absolute A4 count and four invariance metrics against the $\sigma = 3$ reference: Kendall’s τ on the rank order of the Top-10 cities (**T10c**) and Top-10 operators (**T10o**) by certified docked-bike count, plus Jaccard similarity on the Top-10 and Top-50 sets (**J10c/J50c**, **J10o/J50o**). Top-10 city membership is invariant ($J = 1.000$) across the grid; $\tau \geq 0.956$ throughout.

σ	A4 st.	A4 sys.	Share	τ T10c	τ T10o	J10c	J50c	J10o	J50o
2.0	597	89	1.29 %	0.956	0.956	1.000	1.000	1.000	1.000
2.5	543	88	1.17 %	0.956	0.956	1.000	1.000	1.000	1.000
3.0	500	88	1.08 %	ref.	ref.	ref.	ref.	ref.	ref.
3.5	459	82	0.99 %	1.000	1.000	1.000	0.961	1.000	1.000
4.0	415	80	0.90 %	1.000	0.956	1.000	0.961	1.000	1.000

Methodological caveat on the MAD scale. The robust 3σ A4 detector uses $\hat{\sigma} = 1.4826 \cdot \text{MAD}$ as a Gaussian-consistent scale estimator. This assumes an approximately unimodal, isotropic spatial distribution of stations per system. For systems whose deployment is intrinsically anisotropic (transit corridors, riverside linear networks) the MAD over-estimates the effective dispersion on the dominant axis and the detector *under-flags* peripheral stations on that axis; for *multi-modal* deployments (nationwide systems clustering around many city centres; see the *Call a Bike* case in Table 11) the single-centroid construction *over-flags*. The threshold sweep above measures the sensitivity of the detector to its own σ parameter; it does not measure the sensitivity to the underlying geometric assumption, which is a known limitation and the subject of the sub-system-clustering refinement noted in Section 5.

S1 — A4 outlier-detector sensitivity to σ .

S2 — Empirical distribution of the A6 zero-capacity rate. The zero-capacity rate r_s^{A6} across the 301 docked systems with at least 20 stations is reported in Table S2; it underpins the $\tau_{A6} = 1\%$ threshold of Section 2.3.

Table S2: Empirical distribution of the zero-capacity rate r_s^{A6} across the 301 docked systems with at least 20 stations and a reachable `station_information` feed. Free-floating systems, where $c = 0$ is uninformative, are excluded from the A6 base.

Bucket	Systems	Share
$r_s^{A6} = 0\%$ (no $c = 0$)	268	89.0 %
$0 < r_s^{A6} \leq 0.5\%$	7	2.3 %
$0.5 < r_s^{A6} \leq 1\%$	4	1.3 %
$1 < r_s^{A6} \leq 2\%$	10	3.3 %
$2 < r_s^{A6} \leq 5\%$	8	2.7 %
$5 < r_s^{A6} \leq 10\%$	2	0.7 %
$r_s^{A6} > 10\%$	2	0.7 %
Summary statistics		
mean $\bar{r}$	0.36 %	
standard deviation $\hat{\sigma}$	2.41 %	
systems with $r_s^{A6} \geq \tau_{A6} = 1\%$	22	

S3 — Empirical distribution of the A7 null-capacity rate. The NaN-capacity rate ρ_s^{A7} across the 640 globally audited systems with at least 20 stations is reported in Table S3; its sharp bimodality justifies the coarse $\tau_{A7} = 0.5$ threshold of Section 2.4.

Table S3: Empirical distribution of the NaN-capacity rate ρ_s^{A7} across the 640 globally audited systems with at least 20 stations and a reachable `station_information` feed. The corpus is sharply bimodal at the extreme values, which justifies a coarse threshold rather than a distribution-anchored one.

Bucket	Systems	Share
$\rho_s^{A7} = 0\%$ (no NaN)	238	37.2 %
$0 < \rho_s^{A7} < 1\%$	4	0.6 %
$1 \leq \rho_s^{A7} < 10\%$	11	1.7 %
$10 \leq \rho_s^{A7} < 50\%$	22	3.4 %
$50 \leq \rho_s^{A7} < 99\%$	41	6.4 %
$99 \leq \rho_s^{A7} < 100\%$	5	0.8 %
$\rho_s^{A7} = 100\%$ (all NaN)	319	49.8 %
Mass at the two extremes	557	87.0 %

Table S4: Column families of `stations_gold_standard_final.parquet` (46 typed columns). Columns marked $\dagger$ are produced by the audit pipeline; * are derived from external sources. The full per-column schema (name, dtype, source, completeness) ships as a JSON Schema, a Frictionless Data Package and a Croissant manifest (`catalogue/metadata/`). The `audit_confidence` enum is *high* when no flag fires, *medium* for a single A3 or A7 flag, and *low* otherwise.

Family	Cols	Examples / source
Identifiers	5	<code>uid</code> , <code>station_id</code> , <code>system_id</code>
Spatial & administrative	6	<code>lat/lon</code> , <code>city</code> , INSEE commune/region
Station description	4	<code>name</code> , <code>address</code> , <code>capacity</code> , <code>n_stations_system</code>
Audit pipeline outputs $\dagger$	13	<code>station_type</code> , <code>capacity_raw/_audited</code> , <code>flag_A1-A7</code> , <code>audit_confidence</code>
Network geometry $\dagger$	5	intra-/inter-system KNN distances, local densities
Topography*	2	elevation, roughness (IGN BD ALTI)
Cycling infrastructure*	2	cycle-lane linear & share (IGN BD TOPO)
Safety*	1	severe cyclist crashes (ONISR BAAC)
Multimodal access*	2	heavy-transit stops within 300 m (GTFS)
Socio-economic context*	5	INSEE Filosofi income, Gini, car-less share
Modal share*	1	commute-by-bike share (INSEE)
Total	46	

Released catalogue schema (column families).

Reference sources for contextual enrichment.

Per-country breakdown of structural classes A1–A4.

Urban areas most affected by the purging protocol.

Empirical distribution of the A3 capacity-profile ratio.

A4 ablation: legacy versus topology-aware detector.

Table S5: Reference sources used for contextual enrichment. Every listed source contributes at least one column to the released parquet schema (Table S4). Out-of-scope sources considered during prototyping but not retained in the v1.0 release (OpenStreetMap point-of-interest density, FUB cycling barometer score) are documented in the audit report.

Source	Granularity	Variables	Operation
INSEE Filosofi	IRIS / municipality	Median income/CU, local Gini, decile, car-less share	Aggregation
INSEE Recensement	municipality	Share of commute by bike (<code>part_velo_travail</code>)	Aggregation
IGN BD ALTI	DEM raster	Elevation, topographic roughness index	Spatial samp
IGN BD TOPO	polylines	Cycle-infrastructure linear (300 m buffer)	Spatial join
ONISR BAAC (Cerema)	geolocated accident	Severe-crash count (500 m, 5 yr)	Spatial join
National GTFS aggregator	GTFS stops	Heavy-transit stops within 300 m	Spatial join

Table S6: Per-country breakdown of the four structural classes A1–A4 on the MobilityData canonical catalogue (1,509 systems, including the 255 French entries). The macro-region class A5 (17 systems globally), zero-capacity A6 (22) and null-capacity A7 (245 / 97,314 stations) are calibrated on station-count-conditioned sub-corpora and are reported in Tables S2 and S3. Per-class counts are not disjoint; “Flagged” aggregates A1–A5 only. Countries with ≥ 30 audited systems or notable hotspot concentrations are shown. A4 requires both $> 5\%$ of stations and ≥ 5 absolute outside-country stations to flag, to avoid edge-of-coast false positives.

Country	Audited	Reach.	Flagged	A1	A2	A3	A4 geo
France (FR)	255	253	38	17	14	4	3
Germany (DE)	251	243	32	21	4	9	–
United States (US)	175	172	0	–	–	–	–
Poland (PL)	94	91	28	–	3	–	25
Norway (NO)	83	46	2	–	–	–	2
Spain (ES)	69	66	1	–	–	–	1
United Kingdom (GB)	56	54	3	–	1	–	2
Finland (FI)	55	55	14	–	–	–	14
Netherlands (NL)	52	52	1	–	–	–	1
Switzerland (CH)	49	46	15	8	3	3	1
Czech Rep. (CZ)	45	43	25	–	16	12	–
Belgium (BE)	37	34	4	–	–	2	2
Italy (IT)	34	33	0	–	–	–	–
Austria (AT)	27	23	1	–	1	1	–
Sweden (SE)	23	23	12	–	–	–	12
Slovakia (SK)	19	19	16	–	–	–	16
Total (48 countries)	1,509	1,420	204	46	48	33	81

Reproducibility. Per-country counts produced by `scripts/audit_live_systems.py --infer-type` against the 1,509-system MobilityData catalogue; the `--infer-type` flag activates operator-name heuristics (regexes for car-sharing / free-floating brands listed at the top of the script) required to fire A1 and A3 where `station_type` is not declared. Numbers re-fetched later will drift as feeds evolve; the 2026-04 snapshot used here is archived under `experiments/e2_threshold_sensitivity/global_audit_snapshot/` (run `verify_snapshot.py` to reproduce these counts). The sibling `global_audit_results_typed.csv` is a later 2026-05 re-crawl and does not match these figures.

Table S7: Top five French urban areas most affected by the purging protocol. “Raw” is the count of `station_information` entries in the unaudited feed; “Cert. dock” is the count of dock-based stations in the Audit Catalogue. The drop is essentially the share of GBFS entries that the audit reclassifies as free-floating (A3), car-sharing (A1), or filtered (A4–A5). The “Flags driving the drop” column lists the anomaly classes that each agglomeration’s flagged systems trigger.

Urban area	Raw	Cert. dock	Drop	Flags driving the drop
Bordeaux	9,921	225	−97.7 %	A3 (Pony), A1 (Citiz), A7 (Dott, Bird)
Paris	19,872	1,507	−92.4 %	A7 (Dott, Bird), A3 (Pony Paris)
Marseille	2,391	221	−90.8 %	A3 (Pony), A7 (Dott)
Strasbourg	310	39	−87.4 %	A1 (Citiz Grand Est)
Lyon	2,057	452	−78.0 %	A3 (Pony, Bird), A7 (Dott)

Table S8: Empirical distribution of $\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}}$ across two corpora: the 95 audited French systems with non-empty `capacity` (split docked / free-floating; the derivation set) and the 267 globally audited systems (live sweep across 33 countries, `station_type` unknown for many; the out-of-sample set). The threshold $\tau_{A3} = 5$ is preserved in this paper as a conservative cut-off; the global distribution shows that the band $[2, 5]$ is not empty (15 systems) and that a future revision of A3 may benefit from a soft / hard two-tier threshold analogous to A6.

	French corpus (n = 95)		Global (n = 267)
Ratio bucket	Docked	Free-floating	All
[1.00, 1.10]	79	9 [†]	197
[1.10, 2.00]	2	0	24
[2.00, 5.00]	0	0	15
[5.00, 20.00]	0	0	12
[20.00, 50.00]	0	0	9
[50.00, ∞)	0	5	10
Total	81	14	267

[†] Two qualitatively different cases populate the lowest bucket. The 79 docked French systems carry stable positive capacities so $\bar{c}_{\text{profile}}$ and $\bar{c}_{\text{actual}}$ coincide by virtue of *the data*; the 9 free-floating French systems sit there by virtue of *the arithmetic*, because their capacity column is mostly NaN or zero and the ratio degenerates to unity rather than reflecting the conditional-averaging publication choice. The same indistinguishability affects the global “All” column, where we cannot a-priori separate clean docks from degenerate-FF without a per-system audit; the top 10 global systems by ratio are however all known free-floating operators (Pony, nextbike, Voi, Bolt), so the $[5, \infty)$ tail tracks the A3 mechanism across the sweep.

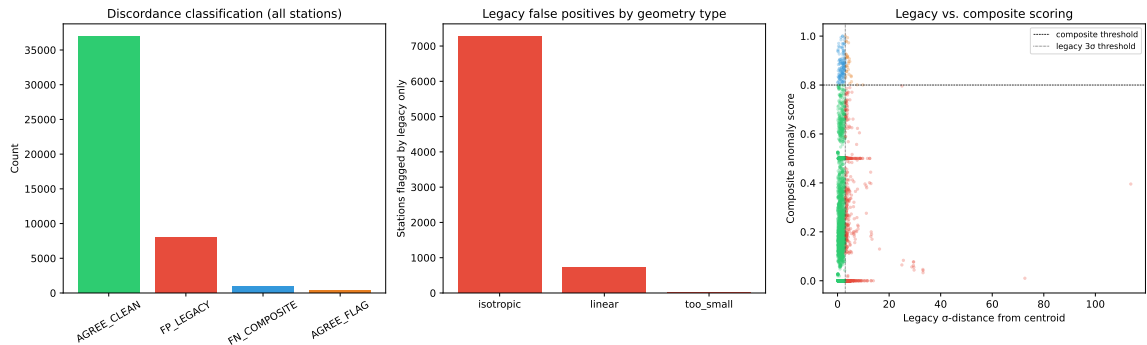


Figure S1: Ablation of the centroid-based A4 detector against the topology-aware composite detector. *Left*: discordance classification across all 46,307 stations. *Centre*: legacy false positives stratified by system geometry type (isotropic vs. linear). *Right*: scatter of legacy σ -distance against composite anomaly score, coloured by discordance class; dashed lines mark the respective detection thresholds.

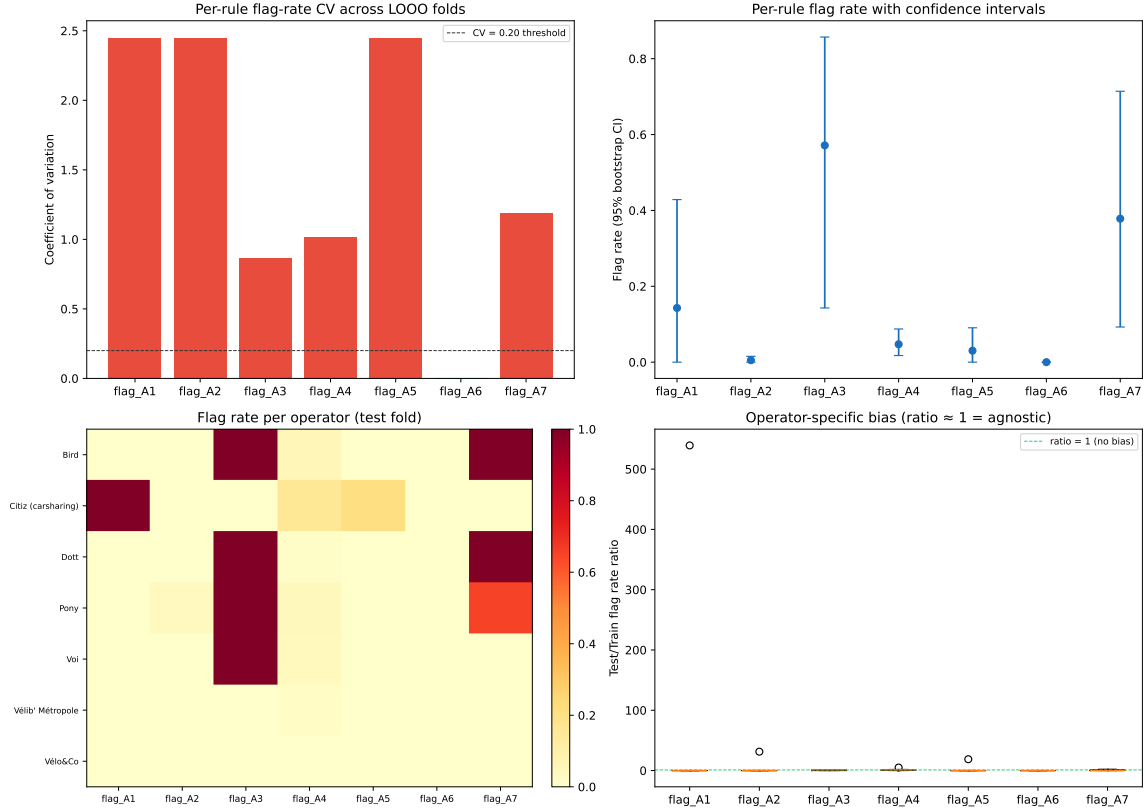


Figure S2: LOOO cross-validation overview. *Top-left*: per-rule coefficient of variation (green = $CV < 0.20$, red = above). *Top-right*: mean flag rate with 95 % bootstrap CI. *Bottom-left*: heatmap of per-operator flag rates on the test fold. *Bottom-right*: test/train flag-rate ratio (ratio ≈ 1 indicates operator-agnostic behaviour).

Leave-one-operator-out cross-validation overview.